

# Aminreza Najafinia

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Ph.D. student in Artificial Intelligence and Soft Computing with research experience in graph neural networks, spatiotemporal modeling, explainable AI, anomaly detection, and machine learning. Experienced in developing data-driven and AI-based systems across academic and industrial projects, with a strong background in programming, optimization, and applied machine learning.

## AI Engineer

Senior Programmer

May 2025 - Present

*Fanavaran Pishrafteh Saqheb*

- Developed an AI-based network security module using UNSW-NB15 and similar datasets with Transformer and Autoencoder models.
- Built a ChatGPT-like system for querying large-scale enterprise data through natural language.
- Developing an AI-assisted Blue Team security tool using Strix for application and network security assessment.

## Frontend Developer

Mid-level Programmer

Apr 2024 - Feb 2025

*Dizbud Gaming Studio*

- Designed the frontend of the landing page and several game pages for the website.
- Worked closely with designers and backend developers to ensure a seamless user experience and responsive designs.

## Machine Learning Freelancer

Freelancer

2023 - Present

*Various Projects*

- Participated in multiple projects and competitions, developing and fine-tuning machine learning models in NLP, image processing, and data analysis.
- Delivered end-to-end solutions, including data preprocessing, feature engineering, model development, and deployment tailored to project requirements.

## EDUCATION

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PH.D. IN ARTIFICIAL INTELLIGENCE AND SOFT COMPUTING

*2026 - Present*

**Amirkabir University of Technology**

Supervisors: [Dr. Zahed Rahmati](#) and [Prof. Amir Goharshady](#)

M.S. IN ARTIFICIAL INTELLIGENCE AND SOFT COMPUTING

*Sep. 2024 - Sep. 2026*

**Amirkabir University of Technology**

GPA: 18.60/20

B.SC. IN COMPUTER SCIENCE

*Nov. 2020 - Jul. 2024*

**Ferdowsi University of Mashhad**

GPA: 18.75/20

ASSISTANT

*2022 - 2024*

**Teaching Assistant, Ferdowsi University of Mashhad**

- Teaching assistant for **Principles of AI**, assisting students with fundamental AI topics including search algorithms (BFS, DFS, A\*), constraint satisfaction problems (CSPs), and problem-solving assignments.

- Teaching assistant for **Principles of Computer and Programming**, assisting students with coding fundamentals and assignments.
- Teaching assistant for **Design of Algorithms**, guiding students through algorithmic problem-solving and coursework.
- Organized and conducted review sessions, graded assignments, and supported students in understanding complex topics.

## RESEARCH & PUBLICATIONS

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### Research Interests

- Machine Learning & Deep Learning
- Graph Neural Networks (GNNs)
- Spatiotemporal Data Analysis
- Explainable AI (XAI)
- Anomaly Detection
- Parameterized Algorithms
- SAT & SMT Solving
- Computational Geometry

### Ph.D. Research

*Supervisors: [Dr. Zahed Rahmati](#) & [Prof. Amir Goharshady](#)*

- Conducting research in Artificial Intelligence and theoretical computer science, with current interests including parameterized algorithms, SAT/SMT solving, computational geometry, graph algorithms, and related algorithmic methods.

### Master's Thesis Research

*Supervisor: [Dr. Zahed Rahmati](#)*

- Investigated dust storm activation mechanisms using graph-based spatiotemporal modeling and explainable AI on hydro-climatic and environmental parameters.
- Master's Thesis Grade: **19.25/20**.

### Publications and Manuscripts

- **Aminreza Najafinia** and **Zahed Rahmati**, “**Three-Year Hydro-Climatic Memory in Dust Activation: A Grouped Temporal Graph Network with Explainable Attention**,” *Scientific Reports, Springer Nature, Under Review*.
- **Aminreza Najafinia** and **Zahed Rahmati**, “**A Spatiotemporal Graph Autoencoder for Early Free-way Traffic Anomaly Detection**,” presented at the *11th International Conference on Signal Processing & Intelligent Systems (ICSPIS)*, December 2025.

## NOTABLE PROJECTS

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- **Face Detection** [Github](#)  
Trained CNN models for gender and age classification based on face images. The workflow includes data loading, preprocessing, image augmentation, model construction, and loss-function customization. Implemented evaluation metrics to monitor performance and fine-tuned hyperparameters to improve model accuracy and generalization.
- **Conversational Search** [GitHub](#)  
Developed a conversational search application integrating a FastAPI backend with OpenAI's GPT for conversational intelligence. Implemented a React frontend for user interaction and utilized Meilisearch for efficient product indexing and search capabilities. Additionally, created a Telegram bot to facilitate user engagement through the Telegram platform.

- **School Scheduling**

[Github](#)

Developed a system for automating school timetable scheduling using optimization techniques. Implemented the system in C# using the IBM ILOG CPLEX optimization library and an OPL (Optimization Programming Language) model.

- **Portfolio Optimization**

[Github](#)

Developed a program to optimize a portfolio of assets by maximizing returns while minimizing risk. Utilized an object-oriented structure for modular and scalable design and integrated optimization techniques such as Tabu Search. Incorporated risk metrics and constraints to simulate real-world financial scenarios.

- **YouTube Data Analysis**

[Github](#)

Analyzed YouTube trending-video data to identify patterns in publishing trends, user interactions, and video performance. The project uses Python libraries for data processing, visualization, and interpretation.

## SKILLS

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### Technical Skills

- Programming Languages: Python, Java, JavaScript, C, C#
- Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn
- Graph Neural Networks: GNNs, Message-Passing Neural Networks, Graph Autoencoders, Temporal Graph Models
- Spatiotemporal Modeling and Time-Series Analysis
- Explainable AI (XAI) and Model Interpretation
- Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Conversational AI
- Anomaly Detection and Representation Learning
- Data Collection, Preprocessing, Feature Engineering, Analysis, and Visualization
- Algorithm Design and Analysis
- Graph Algorithms and Parameterized Algorithms
- SAT/SMT Solving, DPLL/CDCL, and Constraint-Solving Methods
- Optimization Algorithms and Mathematical Optimization
- IBM ILOG CPLEX and Optimization Programming Language (OPL)
- Frontend Development: React, JavaScript, HTML, CSS
- Development Tools: Git, GitHub, Linux, Docker, Jupyter

### Soft Skills

- Mathematical Thinking and Analytical Reasoning
- Problem Solving and Independent Research
- Adaptability and Fast Learning
- Teamwork and Technical Communication
- Familiarity with Scrum and Agile Methodologies